

Core Pure Mathematics 1

Notes · Edexcel A Level Further Mathematics — Core Pure 1 · Chapter 9: Vectors

Chapter 9: Vectors

Why this matters

You already know how to write and manipulate 3D position vectors from A Level Maths. Further Maths builds directly on top of that: this chapter is about describing lines and planes in 3D using vector equations, using the scalar product to find angles and check perpendicularity, and calculating distances between points, lines, and planes. These are genuinely new tools, and together they let you fully describe and analyse 3D space — skills that carry straight through to university-level maths, engineering and physics.

Concept: the vector equation of a line

A line in 3D is fully determined by one point on it and its direction. If $\mathbf{a}$ is the position vector of a known point on the line, and $\mathbf{d}$ is a direction vector parallel to the line, then every point $\mathbf{r}$ on the line satisfies:

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$$

where λ (lambda) is a scalar parameter that varies over all real numbers as we move along the line.

Key vocabulary

Term	Meaning
Direction vector	A vector parallel to a line, giving its orientation (not unique — any scalar multiple works)
Parameter, λ	A scalar that varies to generate every point on a line or plane
Scalar (dot) product	$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = \mathbf{a} \mathbf{b} \cos\theta$
Normal vector, $\mathbf{n}$	A vector perpendicular to a plane
Skew lines	Lines that are neither parallel nor intersecting (only possible in 3D)
Cartesian form of a line	$(x-a_1)/d_1 = (y-a_2)/d_2 = (z-a_3)/d_3$, derived from the vector equation

Worked example 1: finding the vector equation of a line

EXAMPLE 1

Question. Find the vector equation of the line through the points $A(1, 2, -1)$ and $B(3, -1, 4)$.

Step 1. Find a direction vector by subtracting position vectors: $\mathbf{AB} = \mathbf{b} - \mathbf{a} = (3-1, -1-2, 4-(-1)) = (2, -3, 5)$. **(M1 A1)**

Step 2. Choose either point as the fixed point on the line; use A: $a = (1, 2, -1)$. **(B1)**

Step 3. Write the vector equation: $r = (1, 2, -1) + \lambda(2, -3, 5)$. **(A1)**

Answer: $r = (1, 2, -1) + \lambda(2, -3, 5)$.

Exam tip

EXAM TIP

Remember that a vector equation of a line is not unique — using point B instead of A, or doubling the direction vector, gives an equally correct equation. Don't panic if your answer doesn't match the mark scheme exactly; check equivalence by substituting a specific λ value and confirming the resulting point lies on both versions of the line.

Concept: the scalar (dot) product and angles

The scalar product of two vectors a and b is $a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$, and it also equals $|a||b|\cos\theta$, where θ is the angle between them. This gives a direct method for finding angles between two lines (using their direction vectors) or checking perpendicularity: two vectors are perpendicular if and only if their scalar product is zero.

This isn't just a definition — it follows from the cosine rule. In the triangle formed by vectors a , b and $(a - b)$, the cosine rule gives $|a - b|^2 = |a|^2 + |b|^2 - 2|a||b|\cos\theta$. Expanding $|a - b|^2 = (a - b) \cdot (a - b) = |a|^2 - 2(a \cdot b) + |b|^2$ and comparing the two expressions shows $a \cdot b = |a||b|\cos\theta$ directly — the scalar product is really just a shortcut for "the cosine rule, done for you."

Note that only the *direction* vectors are used here — the lines' actual positions in space don't affect the angle between them, so two direction vectors with a zero scalar product mean the lines are perpendicular even if they never actually meet (as with skew lines).

Worked example 2: angle between two lines using the scalar product

EXAMPLE 2

Question. Find the acute angle between the lines with direction vectors $d_1 = (1, 2, 2)$ and $d_2 = (2, -1, 2)$, giving your answer to 1 decimal place.

Step 1. Calculate the scalar product: $d_1 \cdot d_2 = (1 \times 2) + (2 \times -1) + (2 \times 2) = 2 - 2 + 4 = 4$. **(M1 A1)**

Step 2. Find the magnitudes: $|d_1| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$; $|d_2| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3$. **(M1 A1)**

Step 3. Use $\cos\theta = (d_1 \cdot d_2) / (|d_1||d_2|) = 4 / (3 \times 3) = 4/9$. **(M1)**

Step 4. $\theta = \cos^{-1}(4/9) = 63.6^\circ$ (1 d.p.). **(A1)**

Answer: The acute angle between the lines is 63.6° .

Exam tip

EXAM TIP

Remember that the scalar product of direction vectors can give an obtuse angle, but the angle "between two lines" is conventionally reported as acute. If $\cos\theta$ comes out negative, take the

acute angle as 180° minus the calculated value (or, more simply, take the absolute value of the scalar product before dividing).

Concept: the vector equation of a plane

A plane is determined by a point on it and a normal vector $\mathbf{n}$ perpendicular to it. If $\mathbf{a}$ is the position vector of a known point and $\mathbf{r} = (x, y, z)$ is a general point on the plane, then $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$, since the vector from $\mathbf{a}$ to $\mathbf{r}$ must lie in the plane and so be perpendicular to $\mathbf{n}$. Expanding this gives the Cartesian equation of the plane, $n_1x + n_2y + n_3z = d$, where $d = \mathbf{a} \cdot \mathbf{n}$.

Worked example 3: finding the Cartesian equation of a plane

EXAMPLE 3

Question. Find the Cartesian equation of the plane through the point $(2, -1, 3)$ with normal vector $\mathbf{n} = (1, 4, -2)$.

Step 1. Use the plane equation $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$, i.e. $\mathbf{n} \cdot \mathbf{r} = \mathbf{n} \cdot \mathbf{a}$. **(M1)**

Step 2. Calculate $\mathbf{n} \cdot \mathbf{a}$: $(1 \times 2) + (4 \times -1) + (-2 \times 3) = 2 - 4 - 6 = -8$. **(M1 A1)**

Step 3. Write $\mathbf{n} \cdot \mathbf{r}$ using general coordinates (x, y, z) : $1x + 4y - 2z$. **(A1)**

Answer: The Cartesian equation of the plane is $x + 4y - 2z = -8$.

Concept: finding a normal vector using the vector (cross) product

When a plane is defined by three points (or a point and two direction vectors lying in the plane) rather than by a normal vector directly, the normal can be found using the vector (cross) product of two vectors lying in the plane: $\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2$, which is automatically perpendicular to both $\mathbf{d}_1$ and $\mathbf{d}_2$, and therefore to the whole plane.

Worked example 4: finding a plane through three points

EXAMPLE 4

Question. Find the Cartesian equation of the plane through the points $A(1, 0, 0)$, $B(0, 1, 0)$ and $C(0, 0, 1)$.

Step 1. Find two direction vectors in the plane: $\mathbf{AB} = \mathbf{b} - \mathbf{a} = (-1, 1, 0)$; $\mathbf{AC} = \mathbf{c} - \mathbf{a} = (-1, 0, 1)$. **(M1 A1)**

Step 2. Find the normal using the cross product: $\mathbf{n} = \mathbf{AB} \times \mathbf{AC} = ((1 \times 1 - 0 \times 0), (0 \times -1 - (-1) \times 1), ((-1) \times 0 - 1 \times (-1))) = (1, 1, 1)$. **(M1 A1)**

Step 3. Use point A and normal $\mathbf{n} = (1, 1, 1)$: $\mathbf{n} \cdot \mathbf{a} = (1 \times 1) + (1 \times 0) + (1 \times 0) = 1$. **(M1)**

Answer: The Cartesian equation of the plane is $x + y + z = 1$.

Exam tip

EXAM TIP

If your specification includes the cross product formula, learn the determinant-style layout for

computing it reliably — the component-by-component formula is easy to misremember and swap signs on. If the cross product is not required, three simultaneous equations (substituting each point into $ax+by+cz=d$) work just as well and only use skills you already have.

Concept: distance from a point to a plane

The shortest (perpendicular) distance from a point P with position vector p to a plane with equation $n \cdot r = d$ is given by:

$$\text{Distance} = |n \cdot p - d| / |n|$$

Worked example 5: distance from a point to a plane

EXAMPLE 5

Question. Find the distance from the point $(5, 1, -2)$ to the plane $x + 4y - 2z = -8$ (the plane from Worked example 3).

Step 1. Identify $n = (1, 4, -2)$, $d = -8$, and $p = (5, 1, -2)$. **(B1)**

Step 2. Calculate $n \cdot p$: $(1 \times 5) + (4 \times 1) + (-2 \times -2) = 5 + 4 + 4 = 13$. **(M1 A1)**

Step 3. Calculate $n \cdot p - d = 13 - (-8) = 21$. **(M1)**

Step 4. Calculate $|n| = \sqrt{1^2 + 4^2 + (-2)^2} = \sqrt{21}$. **(A1)**

Step 5. Distance $= |21| / \sqrt{21} = 21/\sqrt{21} = \sqrt{21}$. **(A1)**

Answer: The distance is $\sqrt{21}$ units.

Exam tip

EXAM TIP

This formula is easy to apply mechanically but it is easy to forget the absolute value bars, occasionally reporting a negative "distance". Always sanity-check the sign of $n \cdot p - d$, and remember that a distance can never be negative, regardless of which side of the plane the point lies on.

Concept: distance between two skew lines

Two lines that are neither parallel nor intersecting (skew lines, defined above) still have a well-defined shortest distance between them, measured along the direction perpendicular to *both* lines at once. Find a vector n perpendicular to both direction vectors (using the cross product, $n = d_1 \times d_2$), then project the vector joining any point on each line onto this common normal:

$$\text{Distance} = |(a_2 - a_1) \cdot n| / |n|$$

where a_1 and a_2 are position vectors of any points on the two lines. This is the same idea as the point-to-plane distance formula above, applied to the "slab" of parallel planes containing each line.

Chapter summary

Idea	Key point
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Vector equation of a line	$r = a + \lambda d$; not unique (any point on the line, any scalar multiple of the direction)
Scalar product	$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3 = a b \cos\theta$; zero means perpendicular
Angle between lines	Use direction vectors in the scalar product formula; report the acute angle
Vector/Cartesian equation of a plane	$(r-a) \cdot n = 0$, i.e. $n \cdot r = n \cdot a$; expand for the Cartesian form
Normal via cross product	$n = d_1 \times d_2$ for two vectors lying in the plane
Distance point to plane	$ n \cdot p - d / n $

Common exam mistakes

- Forgetting that a vector equation of a line is not unique, and marking a correct alternative answer as wrong.
- Reporting the obtuse angle between two lines instead of the conventional acute angle.
- Sign or component errors when calculating a cross product.
- Forgetting the absolute value in the point-to-plane distance formula, or forgetting to divide by $|n|$.
- Confusing a direction vector (used for lines) with a normal vector (used for planes) — they play very different roles.

You should now be able to

- Write the vector equation of a line given a point and direction, or two points.
- Use the scalar product to find the angle between two lines and to test for perpendicularity.
- Find the vector and Cartesian equations of a plane, given a point and normal, or three points.
- Use the vector (cross) product to find a normal vector to a plane.
- Calculate the shortest distance from a point to a plane.